

MICROSOFT CHALLENGE — MULTIMEDIA PROJECT

Hand Gesture Classification on Edge Devices

Team 14 · Compact gesture classifier for real-world deployment

報告流程 Agenda

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Problem & Scoring
Definition and rules

02

Model Architecture
Dual-Stream Fusion

03

Training & Results
Pipeline and accuracy

Phase 1: Image Branch · Phase 2: Joint Training · Calibration: Threshold Tuning · Live Demo

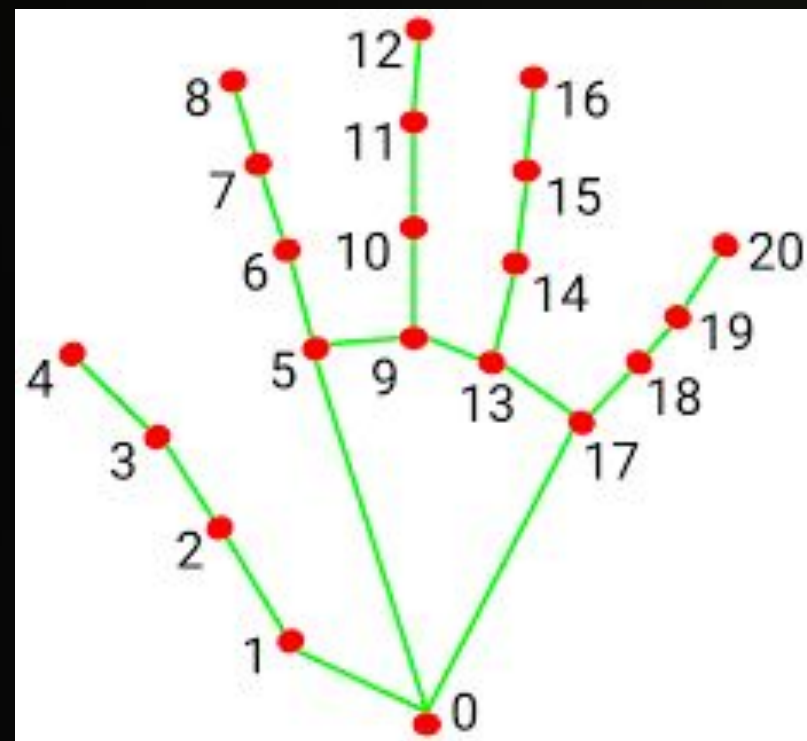
問題定義 Problem Definition

Input Streams

📷 **Cropped RGB Image:** MediaPipe pre-cropped, variable size.

👤 **21 Landmarks:** (x, y) relative coordinates [0, 1].

Constraint: Full-frame images are strictly prohibited.



評分規則 Scoring Rules

Criterion	Points	Formula
Model Size ≤ 10 MB	30	$(10 - \text{size_MB}) \times 3$
Performance (HaGRIDv2)	20	+1 correct, -2 false trigger
Robustness (TA-shot)	40	50 N/A + 50 target images; same scoring
Presentation	30	Live demo + explanation

Key Insight: False trigger = -2 pts | Correct = +1 pt. Conservative N/A rejection is the highest-leverage optimization.

模型架構 **Architecture**

Dual-Stream Fusion

Combining visual texture (CNN) with structural geometry (MLP) for robust classification.

36K

Parameters

0.142

Size (MB)



Image Branch CNN Detail

MobileNet-style DW-Sep CNN

Block = DWConv → BN → ReLU → PWConv → BN → ReLU

Block	In/Out	Stride
Block 1	64→32	2
Block 2	32→16	2
Block 3	16→8	2
Block 4/5	8→8	1



Efficiency Win

DW-sep conv uses **~8–9× fewer** operations than standard convolutions, enabling deployment on extreme edge constraints.

Landmark Branch & Fusion Head



Landmark Branch

Input: 42 floats (21 pts). Encodes **structural geometry** independent of lighting or color via a 2-layer MLP.



Fusion Head

Concatenates both features (128-dim). Uses Dropout (0.3) to prevent co-adaptation between streams.



Complementary

Texture + Structure. Neither alone achieves the same accuracy as the dual-stream approach.

N/A Rejection 3-Layer Defense

Layer 1

Confidence Gate

Rejects predictions if $\max(\text{prob}) < \text{thresh}$. Handles network uncertainty.

Layer 2

Top-2 Margin

Rejects if $1\text{st} - 2\text{nd} < \text{margin}$.
Handles ambiguous tied classes.

Layer 3

Landmark Spread

Rejects if hand span $< 5\%$ of crop.
Filter out degenerate detections.

訓練流程 Pipeline Overview



Phase 1 Image Pretraining

Warm-start Strategy

- ✓ Adam Optimizer, $lr = 1e-3$
- ✓ Cosine Annealing Scheduler
- ✓ **WeightedRandomSampler**: Balances 8× N/A imbalance

86%

BEST VAL ACCURACY

CNN weights frozen-copied for Phase 2

Phase 2 Joint End-to-End

96.7%

VALIDATION ACCURACY

Optimization Tricks

Differential Learning Rates:

- Image Branch: $1e-4$
- Landmark + Fusion: $5e-4$

Weighted Cross-Entropy: N/A class weight $\times$ **1.5 boost** to penalize false triggers.

Calibration Threshold Grid Search

Grid search over **154 combinations** to maximize contest score:

- conf_thresh: 0.30 to 0.95 (14 steps)
- margin_thresh: 0.00 to 0.50 (11 steps)

Scoring Formula: +1 correct, -2 false trigger, 0 N/A.



Heatmap: Contest
Score

數據集 HaGRIDv2 512px

Split	N/A (0)	Target (1-5)	Total
Train	487,094	~106,000	593,703
Val	58,407	~13,500	71,982
Test	99,385	~23,000	122,780



Imbalance Alert

N/A class is **~8× larger** than any target class. Handled via Weighted Sampler and Scoring-Aware Loss.

Data Augmentation TA-Sync

Transform	Parameter
Bbox Jitter	$\pm 10\%$ expansion
Gaussian Blur	σ 0.1–1.5
Rotation	$\pm 15^\circ$
Color Jitter	$\pm 20\%$ intensity

Landmark Sync

Landmarks must rotate with the image:

$$x' = (x - 0.5) \cdot \cos \theta - (y - 0.5) \cdot \sin \theta + 0.5$$

數據結果 Results Summary

We trained 2 runs

Rank	Model Size (30)	Basic Performance (20)	Real World Robustness (40)	Total Score (90)
 1	29.5	19.4	32.5	81.4
 2	29.5	20	31	80.5

Performance

Rank1, 2

Best Contest Score

Efficiency

0.14MB

Total FP32 Model Size

Score

29.57

Size Pts (Max 30)

TEAM 14 PRESENTATION

Live Demo

Showcasing real-time recognition of target gestures and robust N/A rejection of ambiguous poses.

▶ fist · like · ok · one · palm